1 **(a)** Define force.

....

.... [1]

(b) Use the definition in **1(a)** to show that the SI base units of force are $\mathrm{kg\,m\,s^{-2}}$.

[1]

(c) A student is investigating the relationship between the force applied to the piston of a syringe and the velocity of fluid exiting the syringe. The student wishes to use the equation

$$v = \frac{Fk}{mA}$$

where:
v is the velocity of fluid exiting the syringe
F is the force applied to the piston
k is a constant with SI base units $\mathrm{m^2\,s^{-1}}$
m is the mass of the fluid in the syringe
A is the cross-sectional area of the piston.

By considering SI base units on both sides of the equation, state and explain whether this equation is homogeneous.

....

.... [2]

[Total: 4]